

Services Requirements (PROD)

Azure Static Web Apps (Frontend)

Service: Azure Static Web Apps

Plan: Standard

Custom Domain: Enabled

Authentication: Entra ID

Traffic:

- ~30,000–90,000 page views/month

Billing Model: Fixed monthly

Pricing Calculator Inputs:

- Tier: Standard
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Azure App Service (Backend API)

Service: Azure App Service (Linux)

Plan Tier: Standard S2 (*baseline*) / Premium P1v3 (*preferred*)

Region: Primary production region

Instance Count: 2 (minimum)

Autoscale: Enabled

OS: Linux

Runtime: .NET / Node / Python (as applicable)

Capacity (per instance):

- vCPU: ~2
- Memory:
 - S2: ~3.5 GB
 - P1v3: ~8 GB

Traffic:

- ~6,000,000 requests/month
- Peak ~20 RPS
- ~2s execution time

Billing Model: Fixed monthly (App Service Plan)

Pricing Calculator Inputs:

- App Service Plan → Standard S2 (*or Premium P1v3*)
 - Instances: 2
 - Linux OS
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Azure Functions

Service: Azure Functions

Plan: Consumption Plan

Runtime: Python

Region: Same as backend

Execution Model: Event-driven

Usage Estimates (Monthly):

- Total executions: **~7,400 → 15,000+**
 - AI orchestration: 2,000–10,000+
 - Background tasks: ~5,400
- Avg execution duration: 5–40 seconds
- Memory:
 - Background tasks: 512 MB
 - AI orchestration: 1–2 GB

Billing Model: Pay-per-execution

Pricing Calculator Inputs:

- Plan: Consumption
 - Executions: ~10,000/month (safe average)
 - Avg duration: ~20 seconds
 - Memory: 1 GB
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Azure Database for PostgreSQL

Service: Azure Database for PostgreSQL – Flexible Server

Tier: General Purpose

SKU: D4ds v5

vCores: 4

Memory: 16 GB

Traffic:

- ~6M queries/month
- Peak ~20 RPS

Storage:

- Premium SSD
- **50 GB**
- Provisioned IOPS: 0 (auto)

Backups:

- PITR enabled
- 50 GB backup storage
- LRS redundancy

High Availability: Disabled (initial PROD phase)

Billing Model: Compute + storage (hourly)

Pricing Calculator Inputs:

- Tier: General Purpose

- Compute: D4ds v5
 - Storage: 50 GB
 - HA: Disabled
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Azure Storage Account

Service: Azure Blob Storage

Account Type: GPv2

Hierarchical Namespace: Enabled (ADLS Gen2)

Redundancy: ZRS

Access Tier: Hot

Capacity:

- **1,000 GB (1 TB)**

Estimated Monthly Operations:

- Writes: ~2,000
- Reads: ~5,000
- Iterative reads: moderate
- Data write: ~100 GB
- Data retrieval: ~5–10 GB
- Index capacity: 1 TB

Billing Model: Pay-as-you-go

Pricing Calculator Inputs:

- Storage: Blob (Hot)
 - Capacity: 1 TB
 - Redundancy: ZRS
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Azure Key Vault

Service: Azure Key Vault

Tier: Standard

Managed HSM: No

Stored Items:

- Secrets: ~10–20
- Keys: 0–2 (software)
- Certificates: 0–2

Operations:

- ~400–500/month

Billing Model: Per-operation

Pricing Calculator Inputs:

- Tier: Standard
 - Operations: ~500/month
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Azure Monitor (Log Analytics + App Insights)

Services:

- Application Insights
- Log Analytics Workspace

Traffic Context:

- ~6,000,000 backend requests/month

Optimized Ingestion (After Sampling):

- Total logs: **~5–15 GB/month**
 - Analytics Logs: ~4–10 GB
 - Basic Logs: ~1–5 GB

Sampling: 5–10%

Retention: 90 days

Billing Model: Ingestion-based

Pricing Calculator Inputs:

- Analytics Logs: ~8 GB/month
 - Basic Logs: ~3 GB/month
 - Retention: 90 days
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Azure Communication Services (Email)

Service: Azure Communication Services – Email

Billing Model: Pay-as-you-go

Estimated Monthly Volume:

- Typical: ~2,000 emails
- Upper bound modeled: **5,000 emails/month**
- Avg payload size: up to 5 MB

Use Cases:

- Account verification
- OTP / MFA
- Submission status notifications

Pricing Calculator Inputs:

- Emails sent: ~5,000/month
 - Payload size: large (upper bound)
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Azure AI Search

Service: Azure AI Search

Tier: Standard (S1)

Search Units: 1–2

Partitions: 1–2

Replicas: 1–2

Indexed Data (Initial → Mid-term):

- Documents: ~10,000–50,000
- Chunks: ~5–50 million

Query Volume:

- ~6,000–15,000 queries/month

Billing Model: Fixed monthly per Search Unit

Pricing Calculator Inputs:

- Tier: Standard S1
 - Search Units: 1 (*baseline*) or 2 (*HA / throughput*)
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Azure OpenAI Service

Service: Azure OpenAI

Model: GPT-4o-mini

Deployment Type: Standard (On-demand)

Throughput Reservation: None (no PTUs)

Usage Estimates (Monthly):

- Requests:
 - Initial PROD: ~2,000–3,000
 - Mature PROD: 10,000+
- Tokens per request: ~9,500
- Total tokens:
 - Initial: ~20–30M
 - Mature: ~95M+

Billing Model: Token-based consumption

Pricing Calculator Inputs:

- Model: GPT-4o-mini
 - Input tokens: ~70–80%
 - Output tokens: ~20–30%
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Final Notes for PROD Cost Modeling

- All services are **production-grade and conservatively sized**
- Flat-cost services dominate predictable spend
- Usage-based services have **clear caps and monitoring**
- AI costs remain **small relative to core infrastructure**
- Architecture is **national-scale ready without overengineering**